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(54) **AUTHORIZATION CONTROL NETWORK
AND METHOD OF VALIDATING AN
AUTHORIZATION REQUEST**

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CPC **G06Q 20/40** (2013.01); **G06Q 20/24**
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G06Q 20/401; **G06Q 20/405**
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,249,092 B2 * 7/2007 Dunn **G06Q 20/10**
235/379
8,768,801 B1 * 7/2014 Cheatham **G06Q 40/02**
705/40

(Continued)

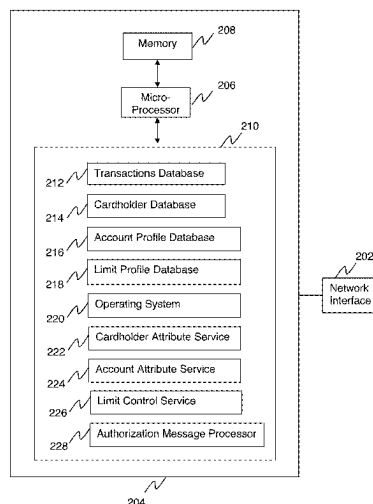
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(57) **ABSTRACT**

An authorization control network includes a limit profile database and at least one computer server. The server(s) receives an account identifier and a datum from a communications device, and associates the datum with the account identifier in a limit profile record of the limit profile database. The server(s) also receives the account identifier and a transaction limit, and associates the transaction limit with the account identifier in the limit profile record. The server(s) then receives from a POS device an authorization request that includes the account identifier and an authorization amount. If the server(s) confirms that the authorization amount does not exceed the transaction limit that is associated with the account identifier in the limit profile record and does not exceed an available credit limit that is associated with the account identifier, the server(s) transmits to the POS device an authorization response that confirms authorization of the authorization amount.

20 Claims, 6 Drawing Sheets



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G06Q 40/03 (2023.01)

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(2013.01); **G06Q 40/03** (2023.01); **G06F**
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(56)

References Cited

U.S. PATENT DOCUMENTS

11,120,447	B2 *	9/2021	Gandhi		G06Q 20/24
11,734,684	B2 *	8/2023	Gandhi		G06Q 20/24
					235/380
2003/0208439	A1 *	11/2003	Rast		G06Q 20/403
					705/38
2012/0265681	A1 *	10/2012	Ross		G06Q 40/02
					705/44
2016/0358250	A1 *	12/2016	Dogin		G06Q 40/03
2019/0156313	A1 *	5/2019	Persaud		G06Q 20/2295

* cited by examiner

Fig. 1

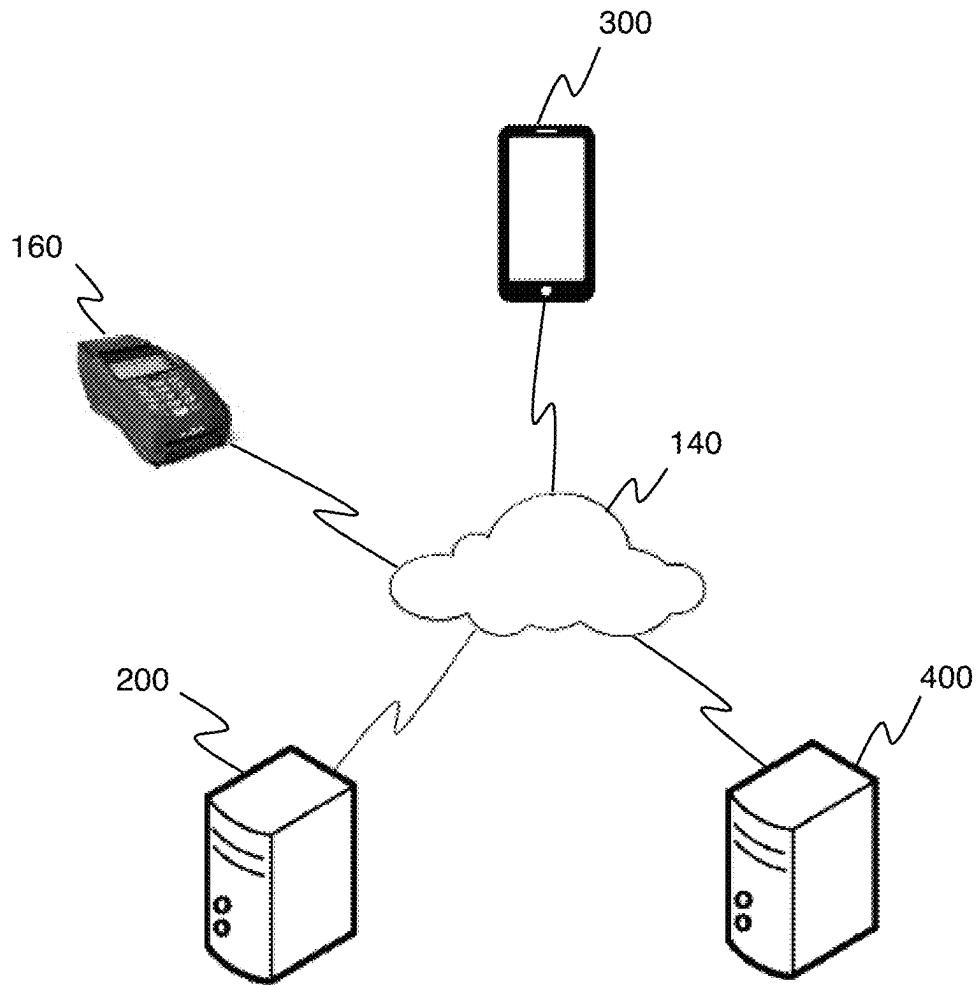


Fig. 2

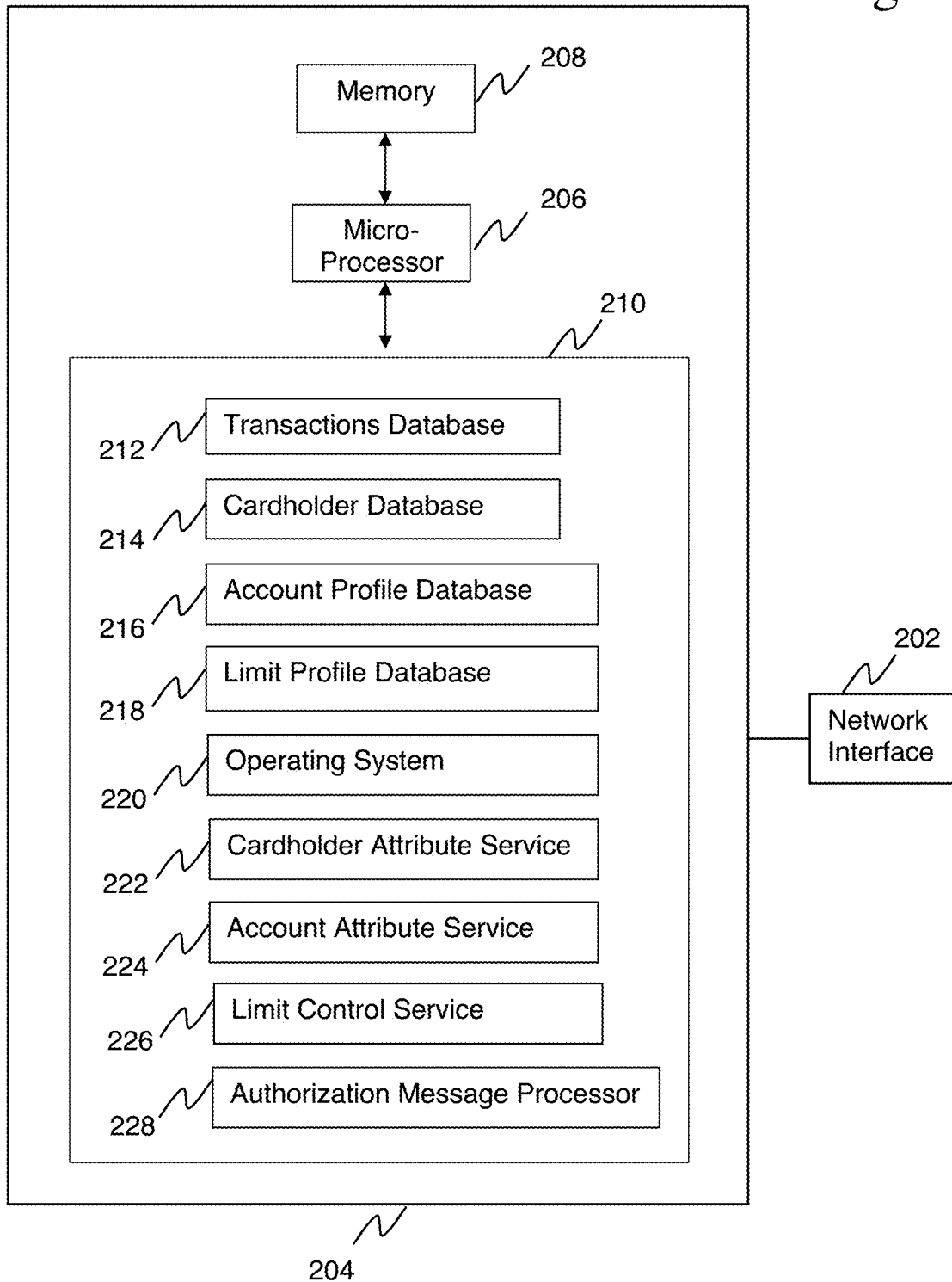


Fig. 3

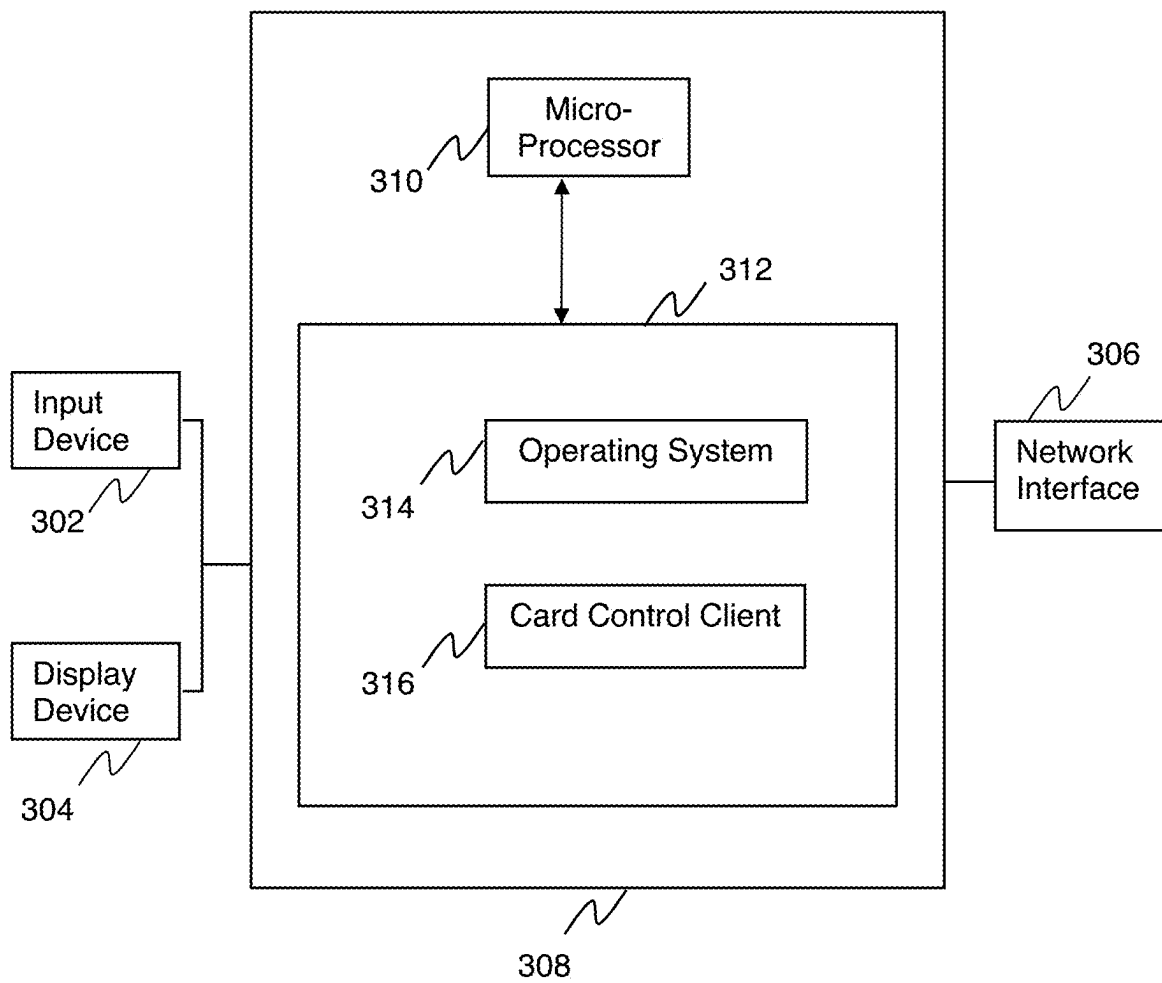
300

Fig. 4

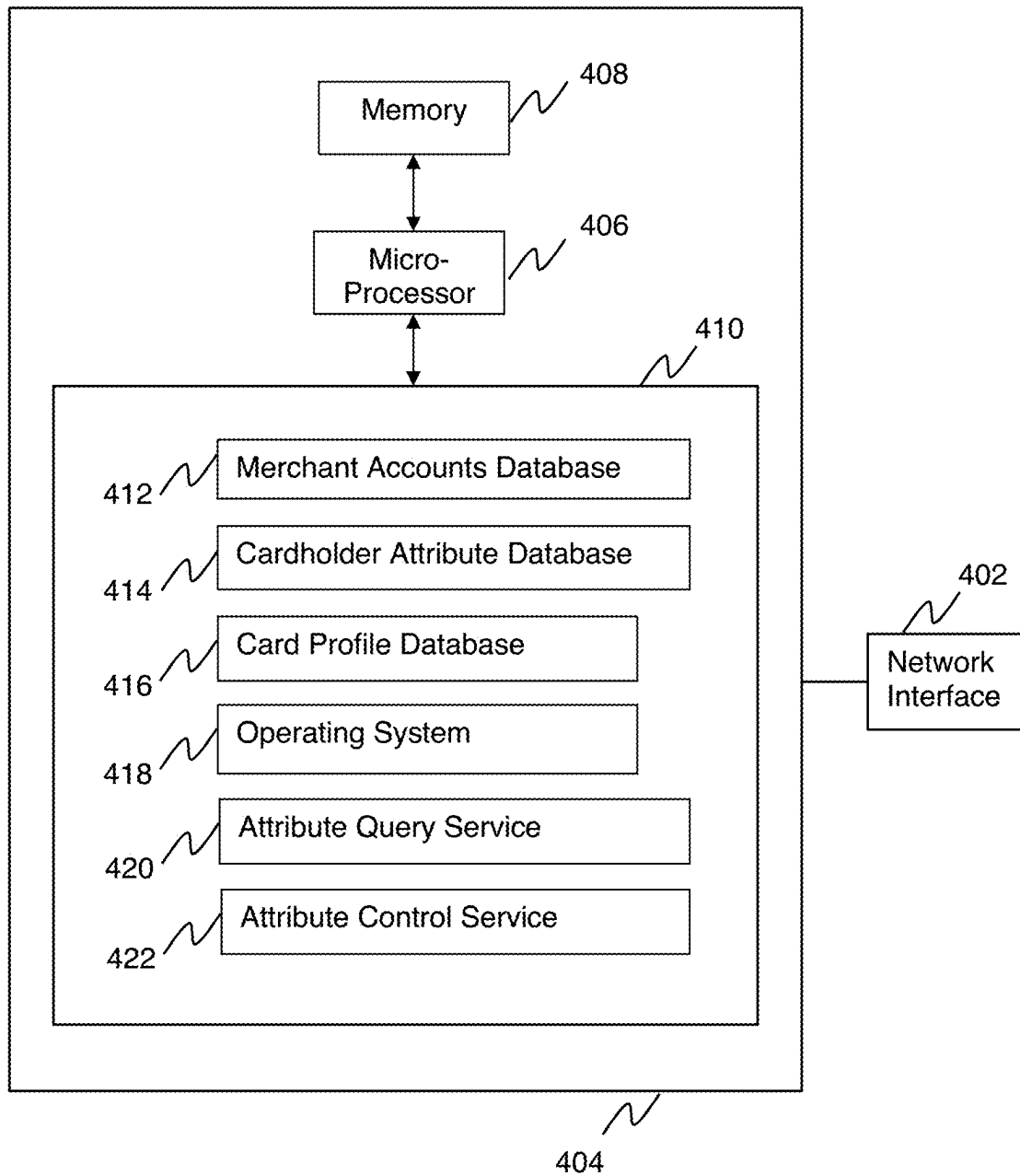


Fig. 5A

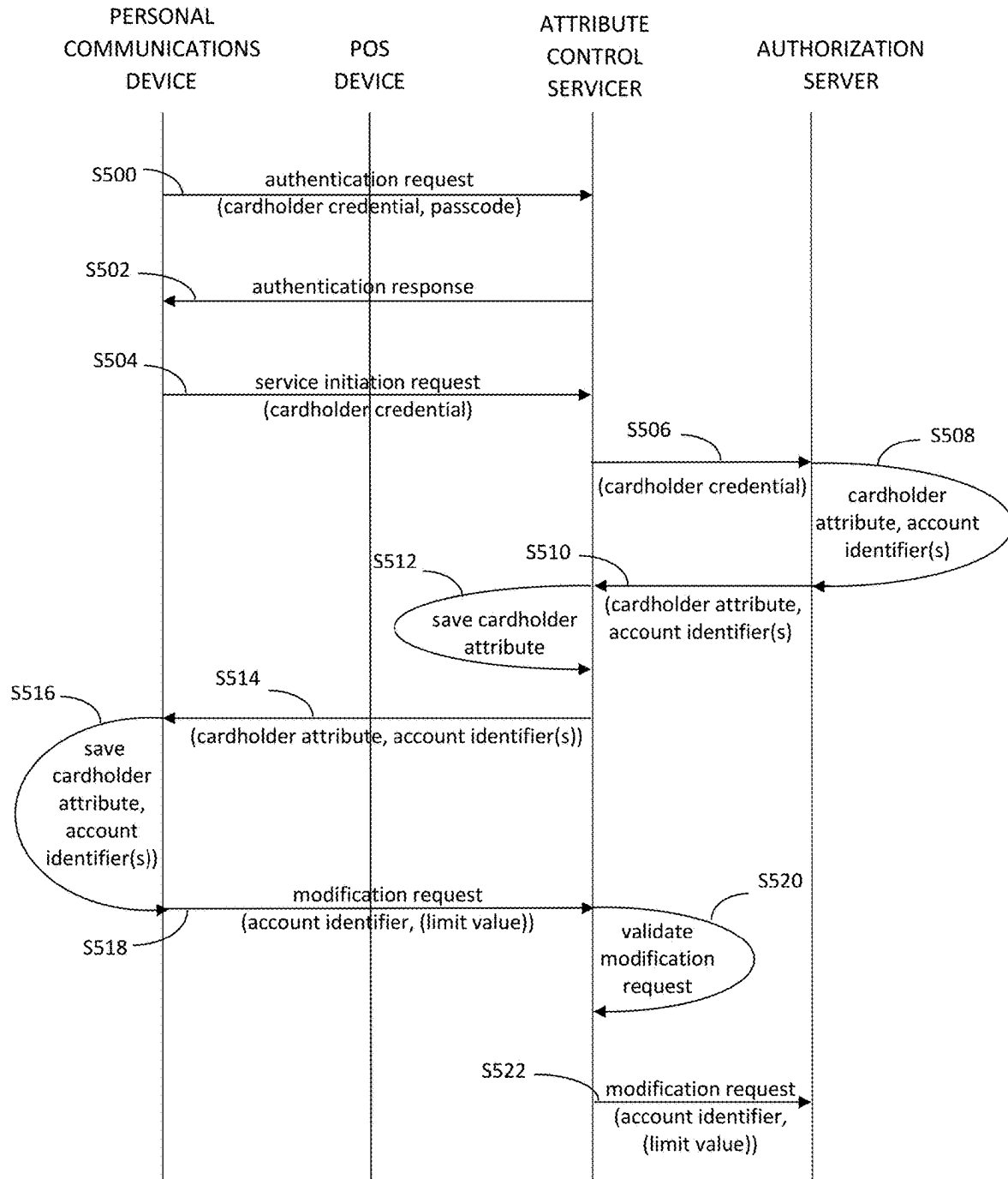
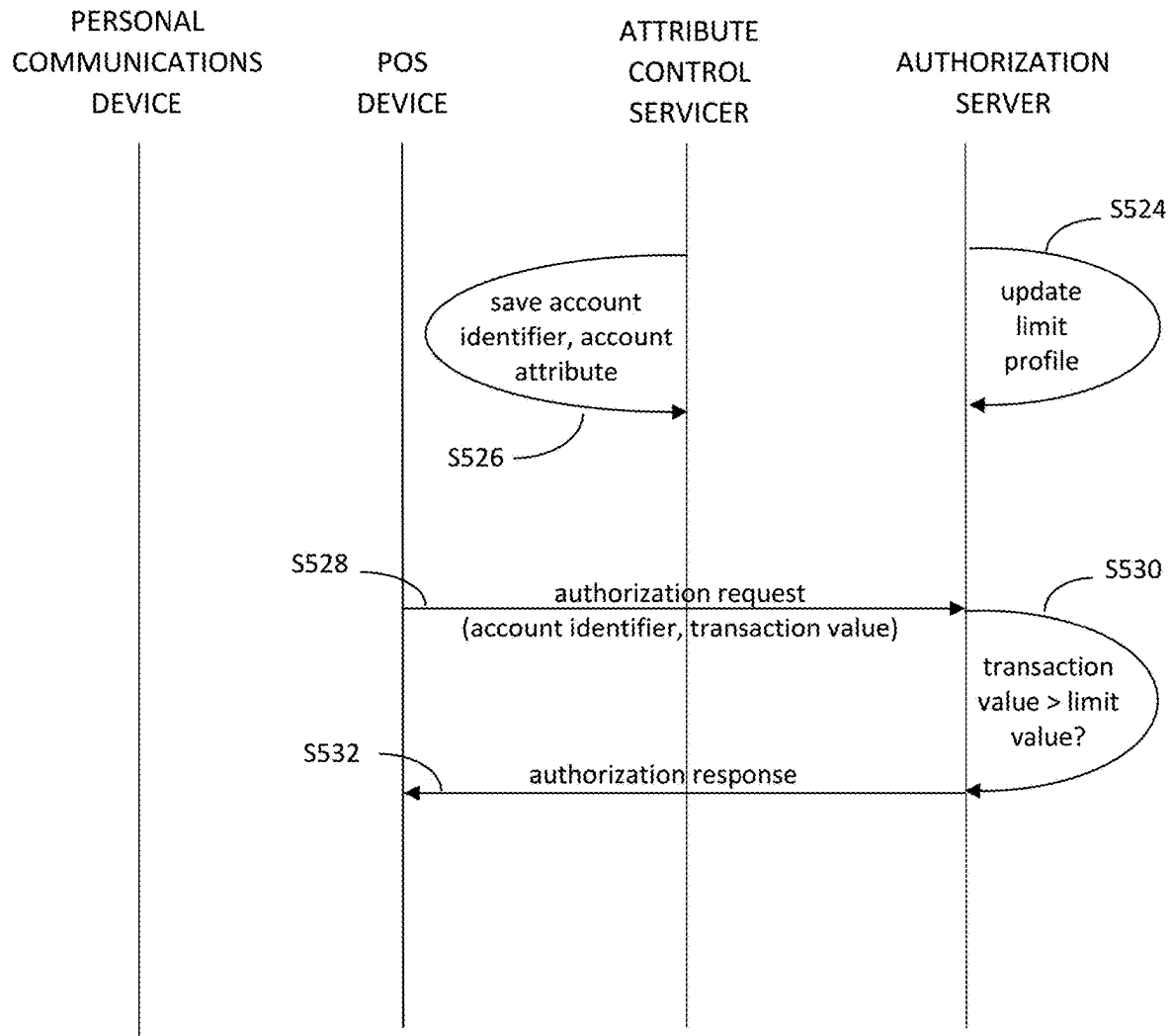


Fig. 5B



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AUTHORIZATION CONTROL NETWORK AND METHOD OF VALIDATING AN AUTHORIZATION REQUEST

RELATED APPLICATIONS

This patent application is a continuation of U.S. patent application Ser. No. 17/400,794, filed Aug. 12, 2021, which is a continuation of U.S. patent application Ser. No. 16/593,579, filed Oct. 4, 2019, the contents of all of which are incorporated herein by reference.

FIELD

This patent application relates to a network and method of authorizing an authorization request.

BACKGROUND

A payment card issuer may provide its customers (cardholders) with payment cards, and the payment card issuer may associate a maximum credit limit value with each payment card, limiting the maximum credit available to cardholders for transactions initiated using their respective payment cards. Therefore, the card issuer may authorize an authorization request received from a point-of-sale (POS) device if the authorization value included in the authorization request does not exceed the maximum credit limit value or if particulars (e.g. merchant, jurisdiction of transaction) of the transaction do not violate fraud threshold parameters implemented by the card issuer.

SUMMARY

This patent application describes an authorization control network, a non-transitory computer-readable medium and an associated method in which at least one computer server receives an authorization request that includes an account identifier and an authorization amount and determines whether the authorization amount exceeds a per-transaction limit value and an available credit limit associated with the account identifier.

In accordance with a first aspect of this disclosure, there is provided an authorization control network that may include a limit profile database and at least one computer server.

The limit profile database includes a plurality of limit profile records. The at least one computer server is configured to receive an account identifier and a datum from a communications device, and associate the datum with the account identifier in a limit profile record of the limit profile database. The at least one computer server is also configured to receive the account identifier and a first transaction limit, and associate the first transaction limit with the account identifier in the limit profile record of the limit profile database.

The at least one computer server is configured to receive from a point-of-sale device an authorization request including the account identifier and an authorization amount, confirm that the authorization amount does not exceed the first transaction limit associated with the account identifier in the limit profile record and does not exceed an available credit limit associated with the account identifier, and transmit to the point-of-sale device an authorization response confirming authorization of the authorization amount.

In accordance with a second aspect of this disclosure, there is provided a method of validating an authorization

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request that involves at least one computer server receiving an account identifier and a datum from a communications device, and associating the datum with the account identifier in a limit profile record of a limit profile database.

The method also involves the at least one computer server receiving the account identifier and a first transaction limit, and associating the first transaction limit with the account identifier in the limit profile record of the limit profile database.

The method further involves the at least one computer server receiving from a point-of-sale device an authorization request including the account identifier and an authorization amount, confirming that the authorization amount does not exceed the first transaction limit associated with the account identifier in the limit profile record and does not exceed an available credit limit associated with the account identifier, and transmitting to the point-of-sale device an authorization response confirming authorization of the authorization amount.

In accordance with a third aspect of this disclosure, there is provided a non-transitory computer-readable medium that includes computer processing instructions which, when executed by a computer server, cause the computer server to receive an account identifier and a datum from a communications device, and associate the datum with the account identifier in a limit profile record of a limit profile database.

The computer processing instructions also cause the computer server to receive the account identifier and a first transaction limit, and associate the first transaction limit with the account identifier in the limit profile record of the limit profile database.

The computer processing instructions further cause the computer server to receive from a point-of-sale device an authorization request including the account identifier and an authorization amount, confirm that the authorization amount does not exceed the first transaction limit associated with the account identifier in the limit profile record and does not exceed an available credit limit associated with the account identifier, and transmit to the point-of-sale device an authorization response confirming authorization of the authorization amount.

In one implementation, the authorization control network includes a card profile database comprising a plurality of card profile records, each storing an account number in association with an account attribute. The server(s) associates the first transaction limit with the account identifier by receiving a cardholder attribute, locating one of the card profile records in the card profile database (wherein the account number in the located card profile record matches the received account identifier), confirming that the account attribute in the located card profile record matches the cardholder attribute, and saving the first transaction limit in the one limit profile record in association with the received account identifier.

In one implementation, the authorization control network includes a cardholder attribute database comprising a plurality of cardholder attribute records, each storing a cardholder identifier in association with a cardholder characteristic.

The server(s) receives the cardholder attribute by receiving a cardholder credential from the communications device, locating one of the cardholder attribute records in the cardholder attribute database (wherein the cardholder identifier in the located cardholder attribute record matches the received cardholder credential), and extracting the cardholder characteristic from the located cardholder attribute

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record (wherein the received cardholder attribute comprises the extracted cardholder characteristic).

In one implementation, the authorization control network includes an account profile database comprising a plurality of account profile records, each storing an account number in association with a primary cardholder credential and a secondary cardholder credential.

The server(s) receives the account identifier from the communications device by receiving a cardholder identifier from the communications device, locating at least one of the account profile records in the account profile database (wherein the primary cardholder credential in each said located account profile record match the received cardholder identifier), downloading each said located account profile record to the communications device (wherein each said downloaded account profile record includes the account number and the associated primary cardholder credential and the associated secondary cardholder credential), and receiving from the communications device one of the account numbers downloaded to the communications device (wherein the received account identifier comprises the received one account number).

In one implementation, the authorization control network includes a card profile database comprising a plurality of card profile records, each storing an account number in association with an account attribute.

The server(s) receives the account identifier, a cardholder attribute and a second transaction limit, and locates one of the card profile records in the card profile database, wherein the account number in the located card profile record match the received account identifier. The server(s) determines that the account attribute in the located card profile record does not match the cardholder attribute, and maintains the first transaction limit in the one limit profile record in association with the received account identifier.

In one implementation, the server(s) receives from a point-of-sale device a second authorization request including the account identifier and a second authorization amount, and determines that the second authorization amount exceeds the first transaction limit associated with the account identifier in the one limit profile record.

Without determining whether the second authorization amount exceeds the available credit limit associated with the account identifier, the server(s) transmits to the point-of-sale device an authorization response advising that the second authorization amount exceeded the first transaction limit.

Since, in accordance with the foregoing aspects of the disclosure, the authorization server updates the limit profile database (i.e. updates the per-transaction limit value in the one limit profile or purges the limit profile from the limit profile database) only after the attribute control server confirms that the cardholder attribute matches the account attribute (i.e. the cardholder requesting the change to the limit profile also caused the one limit profile to be saved in the limit profile database), the security of the technological process of updating the limit profile database is improved in comparison to the state of the art.

BRIEF DESCRIPTION OF THE DRAWINGS

An exemplary authorization control network and method of validating an authorization request will now be described, with reference to the accompanying drawings, in which:

FIG. 1 is a schematic view of an exemplary authorization control network, depicting a point-of-sale device, an authorization server, a personal communications device, and an attribute control server;

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FIG. 2 is a schematic view of the authorization server;

FIG. 3 is a schematic view of the personal communications device;

FIG. 4 is a schematic view of the attribute control server; and

FIGS. 5A and 5B together are a message flow diagram that depicts an exemplary method of validating an authorization request.

DETAILED DESCRIPTION

Authorization Control Network—Overview

FIG. 1 is a schematic view of an authorization control network, denoted generally as 100. As shown, the authorization control network 100 includes a point-of-sale (POS) device 160, an authorization server 200, a personal communications device 300, and an attribute control server 400.

The POS device 160 is typically deployed at a merchant's business premises, and is configured to communicate with the authorization server 200 via a wide area network 140. The POS device 160 may be implemented as an integrated point-of-sale (POS) terminal or as a semi-integrated point-of-sale (POS) terminal, and includes a payment card interface that is configured to communicate with a payment card (e.g. credit card, debit card) that may be interfaced with the POS device 160.

The authorization server 200 may be associated with a financial institution ("cardholder financial institution"), and is configured to communicate with the POS device 160 and the attribute control server 400 via the wide area network 140. As will be discussed in greater detail, the authorization server 200 may maintain (or be in communication with) at least a transactions database that stores particulars of financial transactions initiated using payment cards assigned to respective cardholders, and a limit profile database that stores, for each payment card, a maximum per-transaction limit value for each financial transaction. The authorization server 200 also implements a limit control service that declines authorization for a transaction if the authorization value for the transaction exceeds the specified maximum limit value.

The attribute control server 400 may be associated with a financial institution ("merchant financial institution"), and is configured to communicate with the personal communications device 300 and the authorization server 200 via the wide area network 140. As will be discussed in greater detail, the attribute control server 400 may maintain (or be in communication with) at least a card profile database 416 that indicates, for each payment card, whether the payment card was enrolled in the limit control service by the primary cardholder or the authorized user. The attribute control server 400 also implements an attribute control service that allows a cardholder to modify the maximum per-transaction limit value (or remove a payment card from the limit control service) only if that cardholder enrolled the payment card in the limit control service.

Although the authorization server 200 and the attribute control server 400 are depicted as distinct computer servers in FIG. 1, all or part of the functionality implemented by the attribute control server 400 may instead be implemented by the authorization server 200.

The wide area network 140 may include a wired communications network and a mobile communications network. The wired communications network is configured for the transmission of data across a physical transmission medium. As an example, the wired communications network may be IEEE 802.3 (i.e. Ethernet)-compliant.

The mobile communications network is configured for the transmission of data across a wireless transmission medium. As an example, the mobile communications network may be LTE, WiMax, UMTS, CDMA and/or GSM-compliant. The mobile communications network typically includes at least one wireless base station subsystem (not shown) that may communicate with the personal communications device **300** via a wireless link (where the personal communications device **300** is implemented as a wireless communications device), and may act as a bridge between the wireless links and the wired communications network.

The wide area network **140** also includes a payment network (e.g. VisaNet®, the Mastercard® Network, the Interac® Network) over which confidential payment information (e.g. payment card account numbers, payment card expiry dates) may be securely transmitted between the POS device **160** and the authorization server **200**.

Authorization Server

The authorization server **200** is implemented as a computer server. As shown in FIG. 2, the authorization server **200** includes a network interface **202**, and a data processing system **204** that is in communication with the network interface **202**. The network interface **202** interfaces the authorization server **200** with the wide area network **140**, and allows the authorization server **200** to communicate with the POS device **160** and the attribute control server **400** via the wide area network **140**.

The data processing system **204** includes one or more microprocessors **206**, a volatile computer-readable memory **208** and a non-volatile computer-readable memory **210**. The computer-readable memory **210** may be provided as one or more of a magnetic storage drive and a solid-state drive, and may store a transactions database **212**, cardholder database **214**, an account profile database **216**, and a limit profile database **218**. Alternately, the data processing system **204** may be in communication with the transactions database **212**, the cardholder database **214**, the account profile database **216** and/or the limit profile database **218** (e.g. via the network interface **202**).

The cardholder financial institution may provide (or may authorize a third party to provide) payment cards to customers (“cardholders”) of the cardholder financial institution. Each payment card is configured/encoded with at least a respective account identifier (primary account number or “PAN”). The transactions database **212** includes a plurality of ledgers each associated with a respective one of the transaction account. Each ledger of the transactions database **212** identifies the account identifier of one of the payment cards, particulars (e.g. merchant identifier, authorization amount) of financial transactions initiated using the payment card, and the credit (if any) available using the transaction account.

The cardholder database **214** includes a plurality of records each associated with a respective cardholder. A subset of the cardholders may be related (e.g. family members), and one of the cardholders (“primary cardholder”) of a cardholder subset may have primarily liability to the cardholder financial institution for effecting payment to the cardholder financial institution for all financial transactions initiated using the payment card assigned to the primary cardholder. The primary cardholder may also have primarily liability to the cardholder financial institution for effecting payment to the cardholder financial institution for all financial transactions initiated using the payment cards assigned to the other cardholders (“authorized users”, or “secondary cardholders”) of the cardholder subset. Therefore, each record of the cardholder database **214** includes a cardholder

credential (e.g. userID) that is uniquely associated with the cardholder, and a cardholder attribute field that indicates whether the cardholder is a primary cardholder or an authorized user.

The account profile database **216** includes a plurality of records each associated with a respective transaction account. Each record of the account profile database **216** includes an account identifier of the associated transaction account, a primary cardholder field and an authorized user field. The primary cardholder field identifies the cardholder credential of the primary cardholder associated with the account identifier. The authorized user field identifies the cardholder credential of the authorized user (if any) associated with the account identifier.

The limit profile database **218** includes a plurality of records each associated with a respective limit profile. Each limit profile of the limit profile database **218** identifies at least an account identifier that is uniquely associated with the respective limit profile, and a per-transaction limit value that is associated with the account identifier.

The non-volatile computer-readable memory **210** may also store computer processing instructions which, when copied into the volatile computer-readable memory **208**, and executed by the microprocessor(s) **206** from the volatile computer-readable memory **208**, implement at least an operating system **220**, a cardholder attribute service **222**, an account attribute service **224**, a limit control service **226**, and an authorization message processor **228**.

The operating system **220** allows the authorization server **200** to at least communicate with the POS device **160** and the attribute control server **400** (via the network interface **202**).

The cardholder attribute service **222** is configured to (i) receive a cardholder credential from the attribute control server **400** (via the network interface **202**); (ii) determine the cardholder attribute (from the cardholder attribute field) by querying the cardholder database **214** with the cardholder credential; and (iii) transmit the cardholder attribute to the attribute control server **400** (via the network interface **202**).

The account attribute service **224** is configured to (i) receive a cardholder credential from the attribute control server **400** (via the network interface **202**); (ii) locate all the account identifiers associated with the cardholder credential in the account profile database **216** by querying the primary cardholder field and the authorized user field of the account profile database **216** with the cardholder credential; (iii) generate an account table that includes the located account identifier(s) and identifies, for each located account identifier, the associated authorized user (if any); and (iv) transmit the account table to the attribute control server **400** (via the network interface **202**).

The limit control service **226** is configured to (i) receive from the attribute control server **400**, via the network interface **202**, a modification request that includes at least an account identifier; and (ii) update the limit profile database **218** with the modification request.

For example, the modification request may be provided as an initialization request message that includes a new per-transaction limit value and the account identifier, and the limit control service **226** may be configured to update the limit profile database **218** by confirming that the limit profile database **218** does not include a limit profile that is associated with the account identifier, creating in the limit profile database **218** a new limit profile that includes the account identifier, and saving the new per-transaction limit value in the new limit profile in association with the account identifier.

The modification request may be provided as an update request message that includes an updated per-transaction limit value and the account identifier, and the limit control service 226 may be configured to update the limit profile database 218 by locating the limit profile that is associated with the account identifier in the limit profile database 218, and replacing the current per-transaction limit value in the located limit profile by saving the updated per-transaction limit value in the located limit profile in association with the account identifier.

Alternately, the modification request may be provided as a cancellation request message that includes the account identifier, and the limit control service 226 may be configured to update the limit profile database 218 by locating the limit profile that is associated with the account identifier in the limit profile database 218, and purging the located limit profile from the limit profile database 218.

The authorization message processor 228 is configured to (i) receive from the POS device 160, via the network interface 202, an authorization request that includes an account identifier and an authorization value; (ii) determine the per-transaction limit value associated with the account identifier by querying the limit profile database 218 with the account identifier; (iii) generate an authorization response indicating whether the authorization value exceeds the per-transaction limit value; and (iv) transmit the authorization response to the POS device 160 via the network interface 202.

The authorization response may also indicate whether the total credit available in the transaction account associated with the account identifier is at least equal to the authorization value. Therefore, the authorization message processor 228 may generate the authorization response by also determining whether the authorization value of a financial transaction that is initiated using a payment card exceeds the total credit (if any) available using the associated transaction account.

Personal Communications Device

The personal communications device 300 may be implemented as a wireless communications device, such as a portable digital assistant (PDA), a tablet computer or a smartphone. Alternately, the personal communications device 300 may be implemented as a wired communications device, such as a portable computer or a desktop computer.

As shown in FIG. 3, the personal communications device 300 may include an input device 302, a display device 304, a network interface 306, and a data processing system 308 that is in communication with the input device 302, the display device 304, and the network interface 306.

The input device 302 may be implemented as a keyboard, touchpad, touchscreen or other input device suitable for allowing the operator to input data and/or commands into the personal communications device 300. The display device 304 may be implemented as a liquid crystal display (LCD) panel, plasma display panel, or other display device suitable for displaying information to the operator of the personal communications device 300.

The network interface 306 interfaces the personal communications device 300 with the wide area network 140, and allows the personal communications device 300 to communicate with at least the attribute control server 400 via the wide area network 140.

The data processing system 308 includes a microprocessor 310 and a non-transient computer-readable memory 312. The computer-readable memory 312 may be provided as a FLASH memory, as an example. The computer-readable memory 312 may store computer processing instructions

which, when accessed from the memory 312 and executed by the microprocessor 310, implement at least an operating system 314, and a card control client 316.

The operating system 314 allows the personal communications device 300 to accept user input from the input device 302 and to display information on the display device 304. The operating system 314 also allows the personal communications device 300 to communicate with the attribute control server 400 via the network interface 306.

The card control client 316 is configured to (i) transmit a cardholder credential to the attribute control server 400 (via the network interface 306); (ii) receive from the attribute control server 400 (via the network interface 306) the cardholder attribute that is associated with the cardholder credential in the cardholder database 214; (iii) receive an account table from the attribute control server 400 (via the network interface 306); and (iv) save the cardholder attribute and the account table in the memory 312. The account table includes the account identifier(s) associated with the cardholder credential in the account profile database 216 and identifies, for each account identifier, the associated authorized user (if any).

The card control client 316 is configured to (i) display the received account identifier(s) on the display device 304; (ii) receive from the input device 302 a user selection of one of the account identifier(s) that are displayed on the display device 304; (iii) transmit to the attribute control server 400 (via the network interface 306) a modification request that includes at least the selected account identifier.

If the received cardholder attribute indicates that the cardholder is a primary cardholder, the card control client 316 is configured to prominently render, on the display device 304 (using, for example, bold-faced text), all of the account identifier(s) included in the account table. Therefore, if the cardholder is a primary cardholder, the display device 304 prominently displays the account identifier(s) of the payment card(s) assigned to the primary cardholder, and the account identifier(s) of the payment cards assigned to the authorized user(s) of the primary cardholder's cardholder subset.

Conversely, if the received cardholder attribute indicates that the cardholder is an authorized user, the card control client 316 is configured to (i) locate the account identifier(s) of the payment card(s) assigned to that authorized user by querying the authorized user field of the account table with the cardholder credential of the authorized user, (ii) prominently render the located account identifier(s) on the display device 304 (using, for example, a bold-faced text), and (iii) less prominently render on the display device 304 all the other account identifier(s) included in the account table. Therefore, if the cardholder is an authorized user, the display device 304 prominently displays the account identifier(s) of the payment card(s) assigned to the authorized user, and "greys out" the account identifier(s) of the payment cards assigned to the primary cardholder and all other authorized users of the primary cardholder's cardholder subset.

As discussed, the modification request may be an initialization (or update) request message that includes a per-transaction limit value and the selected account identifier, and the authorization server 200 may receive the initialization (or update) request message from the attribute control server 400 and may be configured to respond to the initialization (or update) request message by saving the per-transaction limit value in the limit profile database 218 in association with the account identifier. Alternately, the modification request may be a cancellation request message that includes the selected account identifier, and the authoriza-

tion server **200** may receive the cancellation request message from the attribute control server **400** and may be configured to respond to the cancellation request message by purging the account identifier and the associated limit value from the limit profile database **218**.

As will be explained, an account identifier prominently displayed on the display device **304** indicates that the attribute control server **400** will accept a modification request in regards to any such account identifier. Conversely, an account identifier that is not prominently displayed on the display device **304** (e.g. displayed using “greyed-out text”) indicates that the attribute control server **400** will not accept a modification request in regards to any such latter account identifier.

Attribute Control Server

The attribute control server **400** is implemented as a computer server. As shown in FIG. **4**, the attribute control server **400** includes a network interface **402**, and a data processing system **404** that is in communication with the network interface **402**. The network interface **402** interfaces the attribute control server **400** with the wide area network **140**, and allows the attribute control server **400** to communicate with the POS device **160** and the attribute control server **400** via the wide area network **140**.

The data processing system **404** includes one or more microprocessors **406**, a volatile computer-readable memory **408** and a non-volatile computer-readable memory **410**. The computer-readable memory **410** may be provided as one or more of a magnetic storage drive and a solid-state drive, and may store a merchant accounts database **412**, a cardholder attribute database **414**, and a card profile database **416**. Alternately, the data processing system **404** may be in communication with the merchant accounts database **412**, the cardholder attribute database **414** and/or the card profile database **416** (e.g. via the network interface **402**).

The merchant accounts database **412** includes a plurality of ledgers each associated with a respective account of a merchant customer (“merchant account”) of the merchant financial institution. Each ledger of the merchant accounts database **412** identifies an account number that is uniquely associated with the respective merchant account, and particulars of various payments made to the merchant account.

The cardholder attribute database **414** includes a plurality of records each associated with a respective cardholder. Each record of the cardholder attribute database **414** includes a cardholder credential that is uniquely associated with the cardholder, and a cardholder attribute field that indicates whether the associated cardholder is a primary cardholder or an authorized user.

The card profile database **416** includes a plurality of records each associated with a respective account identifier. As discussed above, the card control client **316** is configured to transmit to the attribute control server **400** an initialization (or update) request message that includes a per-transaction limit value and a user-selected account identifier, and the authorization server **200** may receive the initialization (or update) request message from the attribute control server **400** and may be configured to respond to the initialization (or update) request message by saving the limit value in the limit profile database **218** in association with the account identifier. Each record of the card profile database **416** includes an account identifier, and an account attribute field that indicates whether the associated account identifier was caused to be saved in the limit profile database **218** by the primary cardholder or the authorized user (if any).

The non-volatile computer-readable memory **410** may also store computer processing instructions which, when

copied into the volatile computer-readable memory **408**, and executed by the microprocessor(s) **406** from the volatile computer-readable memory **408**, implement at least an operating system **418**, an attribute query service **420** and an attribute control service **422**.

The operating system **418** allows the attribute control server **400** to at least communicate with the POS device **160** and the attribute control server **400** (via the network interface **402**).

As discussed, the cardholder attribute service **222** is configured to (i) receive a cardholder credential from the attribute control server **400**; and (ii) transmit to the attribute control server **400** the cardholder attribute that is associated with the cardholder credential in the cardholder database **214**. Further, the account attribute service **224** is configured to (i) receive a cardholder credential from the attribute control server **400**; and (ii) transmit to the attribute control server **400** an account table that includes all the account identifiers associated with the cardholder credential in the account profile database **216** and the respective associated authorized users (if any). Therefore, the attribute query service **420** may receive from the personal communications device **300** a service initiation request that includes the cardholder credential. The attribute query service **420** is configured to respond to the service initiation request by (i) transmitting the cardholder credential to the authorization server **200**, via the network interface **402**, prior to receiving the modification request from the communications device; (ii) receiving from the authorization server **200**, via the network interface **402**, a cardholder attribute that indicates whether the associated cardholder is a primary cardholder or an authorized user; (iii) receiving from the authorization server **200**, via the network interface **402**, an account table that includes the account identifier(s) associated with the cardholder attribute and, for each account identifier, identifies the associated authorized user (if any); and (iv) saving the cardholder attribute in the cardholder attribute database **414** in association with the cardholder credential. The attribute query service **420** may also transmit the cardholder attribute and the account table to the personal communications device **300**, via the network interface **402**.

The attribute control service **422** is configured to (i) receive from the personal communications device **300**, via the network interface **402**, a modification request that includes at least an account identifier; (ii) determine a cardholder attribute from the cardholder credential; (iii) determine an account attribute from the account identifier; (iv) validate the modification request by determining whether the cardholder attribute matches the account attribute; and (v) in accordance with an outcome of the validating, transmit the modification request to the authorization server **200** via the network interface **402**.

The attribute control service **422** is configured to determine the cardholder attribute by querying the cardholder attribute database **414** with the received cardholder credential. The attribute control service **422** is configured to determine the account attribute by locating, in the card profile database **416**, a record that includes the account identifier (included in the modification request), and extracting the account attribute from the account attribute field of the located record. The attribute control service **422** is also configured to validate the modification request by confirming that the cardholder attribute (saved in the cardholder attribute database **414**) matches the account attribute extracted from the card profile database **416**.

As discussed, the limit control service **226** is configured to (i) receive from the attribute control server **400**, via the

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network interface **202**, a modification request that includes at least an account identifier; and (ii) update the limit profile database **218** with the modification request (e.g. save the account identifier in a limit profile in the limit profile database **218**). The attribute control service **422** is configured to transmit the modification request to the authorization server **200** if the card profile database **416** does not have a record that includes the account identifier (i.e. the account identifier has not yet been saved in the limit profile database **218**, or the account identifier has been purged from the limit profile database **218**).

The attribute control service **422** is also configured to transmit the modification request to the authorization server **200** if the cardholder attribute matches the account attribute. Therefore, in effect, the attribute control service **422** transmits a (subsequent) modification request (e.g. including an update request message or a cancellation request message) to the authorization server **200** if the cardholder is a primary cardholder, and the account identifier was caused to be saved in the limit profile database **218** by the primary cardholder (or the cardholder is an authorized user, and the account identifier was caused to be saved in the limit profile database **218** by the authorized user). In this scenario, the attribute control service **422** is also configured to purge, from the card profile database **416**, the record that includes the account identifier, if the modification request includes a cancellation request message.

Conversely, the attribute control service **422** is also configured to ignore a (subsequent) modification request if the cardholder attribute does not match the account attribute. Therefore, in effect, the attribute control service **422** does not transmit a (subsequent) modification request (e.g. including an update request message or a cancellation request message) to the authorization server **200** if the cardholder is a primary cardholder, and the account identifier was caused to be saved in the limit profile database **218** by the authorized user (or the cardholder is an authorized user, and the account identifier was caused to be saved in the limit profile database **218** by the primary cardholder). In this scenario, since the limit control service **226** does not receive the modification request, the limit control service **226** maintains the associated limit profile unchanged.

Method of Validating an Authorization Request

As discussed, the authorization control network **100** implements a method of validating an authorization request. A sample embodiment of the method will be discussed with reference to FIGS. **5A** and **5B**. As will be explained, in this embodiment, at least one computer server (e.g. the authorization server **200**, the attribute control server **400**) receives a cardholder credential and a modification request from a communications device (e.g. the personal communications device **300**). The modification request includes an account identifier.

The computer server(s) (i) determine(s) a cardholder attribute from the cardholder credential, (ii) determine(s) an account attribute by querying a card profile database (e.g. card profile database **416**) with the account identifier, (iii) validate(s) the modification request by determining whether the cardholder attribute matches the account attribute, and (iv) in accordance with an outcome of the validating, update(s) a limit profile database (e.g. the limit profile database **218**) with the modification request. In this embodiment, the card profile database includes at least one record, and one of the records stores the account attribute in association with an account identifier. Further, the limit profile database includes at least one limit profile, and one

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of the limit profiles stores a per-transaction limit value in association with the account identifier.

In this embodiment, the computer server(s) also receive(s) an authorization request from a POS device (e.g. the POS device **160**). The authorization request includes the account identifier and an authorization value. The computer server(s) then (i) determine(s) the per-transaction limit value that is associated with the account identifier by querying the limit profile database with the account identifier, (ii) generate(s) an authorization response, and (iii) transmit(s) the authorization response to the POS device. The authorization response indicates whether the authorization value exceeds the per-transaction limit value.

An example method of validating an authorization request will now be discussed in detail with reference to FIGS. **5A** and **5B**. In the following example, a merchant is in possession of the POS device **160**. The cardholder is in possession of a payment card that is encoded with the account identifier of the cardholder's transaction account. The cardholder is also in possession of the personal communications device **300**.

The attribute control server **400** stores (or is in communication with) a secure cardholder authentication database (not shown) that identifies a set of authentication credentials for each cardholder (e.g. the cardholder's unique cardholder credential and the associated passcode). The transactions database **212** identifies the account identifier of the cardholder's transaction account, particulars of financial transactions initiated using the payment card, and the credit (if any) available using the transaction account.

The cardholder database **214** identifies the cardholder's cardholder credential, and indicates whether the cardholder is a primary cardholder or an authorized user. The account profile database **216** identifies the account identifier of the cardholder's transaction account, and the cardholder credential of the primary cardholder and the cardholder credential of the authorized user (if any) of the transaction account.

However, the account identifier has not yet been enrolled in the limit control service **226** (i.e. the limit profile database **218** does not yet include a limit profile that includes the account identifier and an associated per-transaction limit value). Therefore, the card profile database **416** does not yet include a record that includes the account identifier, and the associated account attribute field (that indicates whether the account identifier was caused to be saved in the limit profile database **218** by the primary cardholder or the authorized user). Similarly, the personal communications device **300** is not yet configured with the account table that identifies the account identifier(s) that are associated with the cardholder's cardholder credential in the account profile database **216** and, for each account identifier, the associated authorized user (if any).

The cardholder attribute database **414** does not yet include a record that includes the cardholder credential, and the associated cardholder attribute that indicates whether the cardholder is a primary cardholder or an authorized user. Therefore, the personal communications device **300** is not yet configured with the cardholder attribute.

(i) Download Cardholder Attribute and Account Table

Prior to attending at the premises of the merchant, the cardholder uses the personal communications device **300** to configure the personal communications device **300** with the cardholder's cardholder identifier and associated account identifier(s). Before the cardholder initiates configuration of the device, however, the cardholder authenticates to the attribute control server **400** by invoking the card control

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client **316**, and entering the cardholder's cardholder credential and passcode into the card control client **316** of the personal communications device **300**. The card control client **316** generates an authentication request (that includes the cardholder credential and passcode) and transmits the authentication request to the attribute control server **400**, at step **S500**.

In response to the authentication request, the attribute control server **400** again authenticates the cardholder with reference to the set of authentication credentials stored in the authentication database. The attribute control server **400** then generates an authentication response that indicates whether the cardholder was successfully authenticated, and transmits the authentication response to the personal communications device **300**, at step **S502**.

The card control client **316** displays the authentication response on the display device **304**. If the authentication response indicates that the cardholder was successfully authenticated, the cardholder then initiates configuration of the personal communications device **300** by using the input device **302** to initiate a configuration service with the attribute control server **400**. The card control client **316** receives the configuration service request, generates a service initiation request that includes the cardholder credential, and transmits the service initiation request to the attribute control server **400**, at step **S504**.

In response to the service initiation request, the attribute query service **420** of the attribute control server **400** transmits the cardholder credential to the authorization server **200**, at step **S506**.

After the authorization server **200** receives the cardholder credential, at step **S508** the cardholder attribute service **222** of the authorization server **200** locates a record in the cardholder database **214** that includes the cardholder credential (e.g. by querying the cardholder database **214** with the cardholder credential), and extracts the cardholder attribute from the cardholder attribute field of the located record.

Similarly, at step **S508** the account attribute service **224** of the authorization server **200** locates all the account identifiers that are associated with the cardholder credential in the account profile database **216** (e.g. by querying the primary cardholder field and the authorized user field of the account profile database **216** with the cardholder credential), and generates an account table that includes the located account identifier(s) and identifies, for each located account identifier, the associated authorized user (if any).

The authorization server **200** transmits the cardholder attribute and the account table to the attribute control server **400**, at step **S510**.

After the attribute control server **400** receives the cardholder attribute and the account table, at step **S512** the attribute query service **420** saves the cardholder attribute in the cardholder attribute database **414** in association with the cardholder credential.

The attribute query service **420** transmits the cardholder attribute and the account table to the personal communications device **300**, at step **S514**. After the personal communications device **300** receives the cardholder attribute and the account table, at step **S516** the card control client **316** saves the cardholder attribute and the account table in the memory **312** of the personal communications device **300**.

(ii) Display Account Identifiers

If the cardholder attribute, received from the attribute control server **400**, indicates that the cardholder is a primary cardholder, the card control client **316** may prominently display, on the display device **304** (using, for example, bold-faced text), all of the account identifier(s) included in

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the account table (i.e. the account identifier(s) of the payment card(s) that are assigned to the primary cardholder, and the account identifier(s) of the payment cards that are assigned to the authorized user(s) of the primary cardholder's cardholder subset).

However, if the cardholder attribute indicates that the cardholder is an authorized user, the card control client **316** may locate the account identifier(s) of the payment card(s) that are assigned to that authorized user (e.g. by querying the authorized user field of the account table with the cardholder credential of the authorized user), and may prominently display the located account identifier(s) on the display device **304** (using, for example, bold-faced text). The card control client **316** may also display the remaining account identifier(s) on the display device **304** (using, for example, greyed out text) (i.e. the account identifier(s) of the payment cards that are assigned to the primary cardholder and all other authorized users of the primary cardholder's cardholder subset). As discussed, the "grey-out" text indicates that the attribute control server **400** will not accept a modification request in regards to any such account identifier.

(iii) Enroll Selected Account Identifier in Limit Control Service

After the card control client **316** displays the account identifier(s) on the display device **304**, the cardholder selects uses the input device **302** to select one of the account identifiers. In this embodiment, the cardholder also uses the input device **302** to enter a new per-transaction limit value, and to request enrollment of the selected account identifier in the limit control service **226**.

The card control client **316** receives the enrollment request, the account identifier and the new limit value, and generates an initialization request message that includes the new per-transaction limit value and the selected account identifier. The card control client **316** then transmits the initialization request message to the attribute control server **400**, at step **S518**.

After the attribute control server **400** receives the initialization request message, at step **S520** the attribute control service **422** validates the initialization request message by determining whether the card profile database **416** has a record that includes the account identifier (e.g. by querying the card profile database **416** with the account identifier). If the attribute control service **422** confirms that the card profile database **416** does not have a record that includes the account identifier (i.e. the selected account identifier is not currently enrolled in the limit profile service **226**), at step **S522** the attribute control service **422** initiates enrollment of the account identifier in the limit profile service **226** by transmitting the initialization request message to the authorization server **200**.

After the authorization server **200** receives the initialization request message, at step **S524** the limit control service **226** of the authorization server **200** updates the limit profile database **218** with the initialization request message (e.g. by confirming that the limit profile database **218** does not include a limit profile that is associated with the account identifier (included in the initialization request message), creating in the limit profile database **218** a new limit profile that includes the account identifier, and saving the new per-transaction limit value (included in the initialization request message) in the new limit profile in association with the account identifier).

Further, if the attribute control service **422** confirms that the card profile database **416** does not have a record that includes the account identifier, at step **S526** the attribute

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control service **422** creates in the card profile database **416** a new record that includes the account identifier, and saves the account attribute (indicating whether the account identifier was enrolled in the limit profile service **226** by the primary cardholder or the authorized user) in the new record in association with the account identifier.

(iv) Modify Limit Control Service

After the account identifier is enrolled in the limit profile service, the cardholder may use the personal communications device **300** to update/modify the per-transaction limit value associated with the account identifier in the limit profile database **218**. Before the cardholder initiates the update service, the cardholder authenticates to the attribute control server **400** by invoking the card control client **316**, and entering the cardholder's cardholder credential and passcode into the card control client **316** of the personal communications device **300**. The card control client **316** generates an authentication request (that includes the cardholder credential and passcode) and transmits the authentication request to the attribute control server **400**, at step **S500**.

In response to the authentication request, the attribute control server **400** again authenticates the cardholder with reference to the set of authentication credentials stored in the authentication database. The attribute control server **400** then generates an authentication response that indicates whether the cardholder was successfully authenticated, and transmits the authentication response to the personal communications device **300**, at step **S502**.

The card control client **316** displays the authentication response on the display device **304**. If the authentication response indicates that the cardholder was successfully authenticated, the cardholder then initiates the update service with the attribute control server **400** by using the input device **302** to select one of the account identifiers, and to request a modification to the limit value associated with the selected account identifier in the limit control service **226**. In this embodiment, the cardholder also uses the input device **302** to enter an updated per-transaction limit value.

The card control client **316** receives the modification request, the account identifier and the updated limit value, and generates an update request message that includes the update limit value and the selected account identifier. The card control client **316** then transmits the update request message to the attribute control server **400**, at step **S518**.

After the attribute control server **400** receives the update request message, the attribute control service **422** determines the cardholder attribute that is associated with the cardholder credential (e.g. by querying the cardholder attribute database **414** with the cardholder credential), and determines the account attribute that is associated with the selected account identifier (e.g. by locating, in the card profile database **416**, a record that includes the account identifier (included in the update request message), and extracting the account attribute from the account attribute field of the located record).

The attribute control service **422** then validates the update request message, at step **S520**, by determining whether the cardholder attribute matches the account attribute. If the attribute control service **422** confirms that the cardholder attribute matches the account attribute (i.e. the cardholder enrolled the selected account identifier in the limit profile service), at step **S522** the attribute control service **422** initiates the update of the associated limit profile in the limit profile service by transmitting the update request message to the authorization server **200**.

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However, if the attribute control service **422** is unable to confirm that the cardholder attribute matches the account attribute (i.e. the cardholder did not enroll the selected account identifier in the limit profile service), the attribute control service **422** does not transmit the update request message to the authorization server **200**. The attribute control service **422** may, however, notify the card control client **316** that the update request has been denied.

After the authorization server **200** receives the update request message, at step **S524** the limit control service **226** of the authorization server **200** updates the limit profile database **218** with the update request message (e.g. by locating the limit profile that is associated with the account identifier (included in the update request message) in the limit profile database **218**, and saving the updated per-transaction limit value (included in the update request message) in the located limit profile in association with the account identifier (in replacement of the current limit value).

(v) Cancel Limit Control Service

After the account identifier is enrolled in the limit profile service, the cardholder may use the personal communications device **300** to cancel the limit profile service for an account identifier (i.e. remove the account identifier from the limit profile database **218**). Before the cardholder initiates the cancellation service, the cardholder authenticates to the attribute control server **400** by invoking the card control client **316**, and entering the cardholder's cardholder credential and passcode into the card control client **316** of the personal communications device **300**. The card control client **316** generates an authentication request (that includes the cardholder credential and passcode) and transmits the authentication request to the attribute control server **400**, at step **S500**.

In response to the authentication request, the attribute control server **400** again authenticates the cardholder with reference to the set of authentication credentials stored in the authentication database. The attribute control server **400** then generates an authentication response that indicates whether the cardholder was successfully authenticated, and transmits the authentication response to the personal communications device **300**, at step **S502**.

The card control client **316** displays the authentication response on the display device **304**. If the authentication response indicates that the cardholder was successfully authenticated, the cardholder then initiates the cancellation service with the attribute control server **400** by using the input device **302** to select one of the account identifiers, and to request a removal of the selected account identifier from the limit control service **226**.

The card control client **316** receives the cancellation request and the account identifier, and generates a cancellation request message that includes the selected account identifier. The card control client **316** then transmits the cancellation request message to the attribute control server **400**, at step **S518**.

After the attribute control server **400** receives the cancellation request message, the attribute control service **422** determines the cardholder attribute that is associated with the cardholder credential (e.g. by querying the cardholder attribute database **414** with the cardholder credential), and determines the account attribute that is associated with the selected account identifier (e.g. by locating, in the card profile database **416**, a record that includes the account identifier (included in the update request message), and extracting the account attribute from the account attribute field of the located record).

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The attribute control service **422** then validates the update request message, at step **S520**, by determining whether the cardholder attribute matches the account attribute. If the attribute control service **422** confirms that the cardholder attribute matches the account attribute (i.e. the cardholder enrolled the selected account identifier in the limit profile service), at step **S522** the attribute control service **422** initiates the cancellation of the limit profile service for the account identifier by transmitting the cancellation request message to the authorization server **200**.

However, if the attribute control service **422** is unable to confirm that the cardholder attribute matches the account attribute (i.e. the cardholder did not enroll the selected account identifier in the limit profile service), the attribute control service **422** does not transmit the cancellation request message to the authorization server **200**. The attribute control service **422** may, however, notify the card control client **316** that the cancellation request has been denied.

After the authorization server **200** receives the cancellation request message, at step **S524** the limit control service **226** of the authorization server **200** updates the limit profile database **218** with the cancellation request message (e.g. by locating the limit profile that is associated with the account identifier (included in the cancellation request message) in the limit profile database **218**, and purging the located limit profile from the limit profile database **218**).

Further, if the attribute control service **422** confirms that the cardholder attribute matches the account attribute, at step **S526** the attribute control service **422** locates the record that is associated with the account identifier in the card profile database **416**, and purges the located record from the card profile database **416**.

Since successful completion of the cancellation service causes the account identifier to be removed from the card profile database **416** (and the limit profile database **218**), another cardholder can subsequently enroll the same account identifier in the limit profile service (and modify the per-transaction limit value associated with the account identifier in the limit profile database **218**).

(vi) Generate Authorization Request

After the account identifier is enrolled in the limit profile service, the cardholder may attend at the premises of a merchant, and may use the merchant's POS device **160** to effect payment for a financial transaction initiated with the merchant. The merchant may input an authorization value (e.g. required payment amount for the financial transaction) into the POS device **160**, and the cardholder may initiate payment of the authorization value by interfacing the cardholder's payment card with the POS device **160**.

The POS device **160** reads the account identifier from the payment card, and generates an authorization request message that includes at least the merchant identifier of the merchant, the account identifier and the authorization value. The POS device **160** then transmits the authorization request message to the authorization server **200** via the payment network of the wide area network **140**, at step **S528**.

After the authorization server **200** receives the authorization request message from the POS device **160**, the authorization server **200** determines whether the authorization request message can be authorized by (i) determining whether the transaction value specified in the authorization request message exceeds the per-transaction limit value associated with the account identifier (if any), and (ii) determining whether the transaction value exceeds the maximum credit available in the associated transaction account.

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Therefore, at step **S530** the authorization message processor **228** of the authorization server **200** determines the per-transaction limit value (if any) associated with the account identifier (e.g. by querying the limit profile database **218** with the account identifier), and determines whether the authorization value exceeds the per-transaction limit value. If the authorization message processor **228** determines that the financial transaction cannot be authorized because the authorization value exceeds the per-transaction limit value, the authorization message processor **228** may generate an authorization response that indicates that the authorization value exceeded the limit value.

However, if the authorization message processor **228** determines that the authorization value does not exceed the per-transaction limit value, the authorization message processor **228** determines from the account identifier whether the transaction (in an amount equal to the authorization value) can be authorized (i.e. determines from the transactions database **212** whether the credit available in the transaction account associated with the account identifier is at least equal to the authorization value). If the authorization message processor **228** determines that the financial transaction cannot be authorized because the authorization value exceeds the credit available in the transaction account, the authorization message processor **228** may generate an authorization response that indicates that the authorization value exceeded the available credit.

However, if the authorization server **200** determines that the transaction can be authorized, the authorization server **200** updates the transactions database **212** with particulars of the transaction (e.g. merchant, transaction value) and generates an authorization response message indicating that the transaction was authorized. After the authorization server **200** generates the authorization response message, the authorization server **200** transmits the authorization response message to the POS device **160** via the payment network, at step **S532**.

After the POS device **160** receives the authorization response message from the authorization server **200**, the POS device **160** determines from the authorization response message whether the transaction was authorized. If the POS device **160** determines that the transaction was authorized, the POS device **160** notifies the customer that the transaction was authorized. Otherwise, the POS device **160** notifies the customer that the transaction was declined.

The invention claimed is:

1. An authorization control network comprising:
 - a limit profile database comprising a plurality of limit profile records; and
 - at least one computer server configured with computer processing instructions, wherein the computer processing instructions cause the at least one computer server to:
 - receive an account identifier and a first transaction limit, and associate the first transaction limit with the account identifier in one of the limit profile records of the limit profile database,
 - receive from a point-of-sale device an authorization request including the account identifier and an authorization amount,
 - confirm that the authorization amount does not exceed the first transaction limit associated with the account identifier in the one limit profile record and does not exceed an available credit limit associated with the account identifier, and

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transmit to the point-of-sale device an authorization response confirming authorization of the authorization amount.

2. The authorization control network according to claim 1, further comprising:

a card profile database comprising a plurality of card profile records each storing an account number in association with an account attribute,

wherein the computer processing instructions cause the at least one computer server to associate the first transaction limit with the account identifier by:

receiving a cardholder attribute,

locating one of the card profile records in the card profile database, the account number in the located card profile record matching the received account identifier,

confirming that the account attribute in the located card profile record matches the cardholder attribute, and

saving the first transaction limit in the one limit profile record in association with the received account identifier.

3. The authorization control network according to claim 2, wherein the computer processing instructions cause the at least one computer server to save the first transaction limit in the one limit profile record by:

replacing, with the first transaction limit, a datum associated with the received account identifier in the one limit profile record.

4. The authorization control network according to claim 2, further comprising:

a cardholder attribute database comprising a plurality of cardholder attribute records each storing a cardholder identifier in association with a cardholder characteristic,

wherein the computer processing instructions cause the at least one computer server to receive the cardholder attribute by:

receiving a cardholder credential from a communications device,

locating one of the cardholder attribute records in the cardholder attribute database, the cardholder identifier in the located cardholder attribute record matching the received cardholder credential, and

extracting the cardholder characteristic from the located cardholder attribute record, the received cardholder attribute comprising the extracted cardholder characteristic.

5. The authorization control network according to claim 1, further comprising:

an account profile database comprising a plurality of account profile records each storing an account number in association with a primary cardholder credential and a secondary cardholder credential,

wherein the computer processing instructions cause the at least one computer server to receive the account identifier by:

receiving a cardholder identifier from a communications device,

locating at least one of the account profile records in the account profile database, the primary cardholder credential in each said located account profile record matching the received cardholder identifier,

downloading each said located account profile record to the communications device, each said downloaded account profile record including the account number and the associated primary cardholder credential and the associated secondary cardholder credential, and

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receiving from the communications device one of the account numbers downloaded to the communications device, the received account identifier comprising the received one account number.

6. The authorization control network according to claim 1, further comprising:

a card profile database comprising a plurality of card profile records each storing an account number in association with an account attribute,

wherein the computer processing instructions further cause the at least one computer server to:

receive the account identifier, a cardholder attribute and a second transaction limit,

locate one of the card profile records in the card profile database, the account number in the located card profile record matching the received account identifier,

determine that the account attribute in the located card profile record does not match the cardholder attribute, and

maintain the first transaction limit in the one limit profile record in association with the received account identifier.

7. The authorization control network according to claim 1, further comprising:

a transactions database comprising a plurality of transaction records each storing an account number in association with an available credit amount,

wherein the computer processing instructions cause the at least one computer server to confirm that the authorization amount does not exceed the available credit limit by:

locating one of the transaction records in the transactions database, the account number in the located transaction record matching the received account identifier, and

confirming that the authorization amount does not exceed the available credit amount in the located transaction record, the available credit limit comprising the available credit amount in the located one transaction record.

8. A method of validating an authorization request comprising at least one computer server:

receiving an account identifier and a first transaction limit, and associating the first transaction limit with the account identifier in a limit profile record of a limit profile database,

receiving from a point-of-sale device an authorization request including the account identifier and an authorization amount,

confirming that the authorization amount does not exceed the first transaction limit associated with the account identifier in the limit profile record and does not exceed an available credit limit associated with the account identifier, and

transmitting to the point-of-sale device an authorization response confirming authorization of the authorization amount.

9. The method according to claim 8, wherein the associating the first transaction limit with the account identifier comprises the at least one computer server:

receiving a cardholder attribute,

locating a card profile record in a card profile database, the located card profile record storing an account number in association with an account attribute, the account number in the located card profile record matching the received account identifier,

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confirming that the account attribute in the located card profile record matches the cardholder attribute, and saving the first transaction limit in the limit profile record in association with the received account identifier.

10. The method according to claim 9, wherein the saving the first transaction limit in the limit profile record comprises the at least one computer server:

replacing, with the first transaction limit, a datum associated with the received account identifier in the limit profile record.

11. The method according to claim 9, wherein the receiving the cardholder attribute comprises the at least one computer server:

receiving a cardholder credential from a communications device,

locating a cardholder attribute record in a cardholder attribute database, the located cardholder attribute record storing a cardholder identifier in association with a cardholder characteristic, the cardholder identifier in the located cardholder attribute record matching the received cardholder credential, and

extracting the cardholder characteristic from the located cardholder attribute record, the received cardholder attribute comprising the extracted cardholder characteristic.

12. The method according to claim 8, wherein the receiving the account identifier comprises the at least one computer server:

receiving a cardholder identifier from a communications device,

locating at least one account profile record in an account profile database, each said located account profile record storing an account number in association with a primary cardholder credential and a secondary cardholder credential, the primary cardholder credential in each said located account profile record matching the received cardholder identifier,

downloading each said located account profile record to the communications device, each said downloaded account profile record including the account number, the associated primary cardholder credential and the associated secondary cardholder credential, and

receiving from the communications device one of the account numbers downloaded to the communications device, the received account identifier comprising the received one account number.

13. The method according to claim 8, further comprising the at least one computer server:

receiving the account identifier, a cardholder attribute and a second transaction limit,

locating a card profile record in a card profile database, the located card profile record storing an account number in association with an account attribute, the located card profile record storing an account number in association with an account attribute, the account number in the located card profile record matching the received account identifier,

determining that the account attribute in the located card profile record does not match the cardholder attribute, and

maintaining the first transaction limit in the limit profile record in association with the received account identifier.

14. The method according to claim 8, wherein the confirming that the authorization amount does not exceed the available credit limit comprises the at least one computer server:

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locating a transaction record in a transactions database, the located transaction record storing an account number in association with an available credit amount, the account number in the located transaction record matching the received account identifier, and

confirming that the authorization amount does not exceed the available credit amount in the located transaction record, the available credit limit comprising the available credit amount in the located transaction record.

15. A tangible non-transient computer-readable medium storing computer processing instructions which, when executed by a computer server, cause the computer server to:

receive an account identifier and a first transaction limit, and associate the first transaction limit with the account identifier in a limit profile record of a limit profile database,

receive from a point-of-sale device an authorization request including the account identifier and an authorization amount,

confirm that the authorization amount does not exceed the first transaction limit associated with the account identifier in the limit profile record and does not exceed an available credit limit associated with the account identifier, and

transmit to the point-of-sale device an authorization response confirming authorization of the authorization amount.

16. The computer-readable medium according to claim 15, wherein the computer processing instructions cause the computer server to associate the first transaction limit with the account identifier by:

receiving a cardholder attribute,

locating a card profile record in a card profile database, the located card profile record storing an account number in association with an account attribute, the account number in the located card profile record matching the received account identifier,

confirming that the account attribute in the located card profile record matches the cardholder attribute, and saving the first transaction limit in the limit profile record in association with the received account identifier.

17. The computer-readable medium according to claim 16, wherein the computer processing instructions cause the computer server to receive the cardholder attribute by:

receiving a cardholder credential from a communications device,

locating a cardholder attribute record in a cardholder attribute database, the located cardholder attribute record storing a cardholder identifier in association with a cardholder characteristic, the cardholder identifier in the located cardholder attribute record matching the received cardholder credential, and

extracting the cardholder characteristic from the located cardholder attribute record, the received cardholder attribute comprising the extracted cardholder characteristic.

18. The computer-readable medium according to claim 15, wherein the computer processing instructions cause the computer server to receive the account identifier by:

receiving a cardholder identifier from a communications device,

locating at least one account profile record in an account profile database, each said located account profile record storing an account number in association with a primary cardholder credential and a secondary cardholder credential, the primary cardholder credential in

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each said located account profile record matching the received cardholder identifier,
 downloading each said located account profile record to the communications device, each said downloaded account profile record including the account number, the associated primary cardholder credential and the associated secondary cardholder credential, and
 receiving from the communications device one of the account numbers downloaded to the communications device, the received account identifier comprising the received one account number.

19. The computer-readable medium according to claim 15, wherein the computer processing instructions further cause the computer server to:

- receive the account identifier, a cardholder attribute and a second transaction limit,
- locate a card profile record in a card profile database, the located card profile record storing an account number in association with an account attribute, the located card profile record storing an account number in association with an account attribute, the account number in the located card profile record matching the received account identifier,

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determine that the account attribute in the located card profile record does not match the cardholder attribute, and

maintain the first transaction limit in the limit profile record in association with the received account identifier.

20. The computer-readable medium according to claim 15, wherein the computer processing instructions cause the computer server to confirm that the authorization amount does not exceed the available credit limit by:

- locating a transaction record in a transactions database, the located transaction record storing an account number in association with an available credit amount, the account number in the located transaction record matching the received account identifier, and
- confirming that the authorization amount does not exceed the available credit amount in the located transaction record, the available credit limit comprising the available credit amount in the located transaction record.

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